Quantifying the Sub-Optimality of zCDP Accounting under Heterogeneous Composition

Abstract

We study the gap between the standard zCDP composition bound $\rho = \sum_i \rho_i$ and the exact privacy curve of a heterogeneous composition $M = (M_1, \ldots, M_n)$ of additive symmetric noise mechanisms. Working with the Rényi-divergence gap functional $\Gamma_{n,\nu}(\alpha) := \rho\alpha - D_\alpha(P\|Q)$, we derive matching upper and lower bounds expressed in terms of: (i) the curvature of the per-mechanism Rényi divergence $\kappa_i(\alpha) := \rho_i\alpha - D_\alpha(P_i\|Q_i)$, (ii) the allocation $(\rho_1, \ldots, \rho_n)$, and (iii) the noise family ν. The qualitative picture is: Gaussian noise is the unique zero of $\Gamma_{n,\nu}$ (zCDP is tight), and for any non-Gaussian symmetric noise satisfying mild regularity the gap grows at least linearly in n at the worst-case allocation and is bounded below by a quadratic form in $(\rho_1, \ldots, \rho_n)$ controlled by the fourth cumulant of ν.

1 Setup and Assumptions

We consider mechanisms of the form $M_i(x) = q_i(x) + Z_i$ where $q_i : \mathcal{X} \to \mathbb{R}$ is a query with global sensitivity Δ_i and Z_i has density ν_{σ_i} from a scale family $\{\nu_\sigma\}_{\sigma>0}$ with $\nu_\sigma(z) = \sigma^{-1}\nu_1(z/\sigma)$. We write $\nu := \nu_1$ for the base density. The mechanisms are run independently. Fix two neighboring datasets x, x' and let $\mu_i := q_i(x) - q_i(x') \in [-\Delta_i, \Delta_i]$. Define

$$P_i := \nu_{\sigma_i}(\cdot - \mu_i), \quad Q_i := \nu_{\sigma_i}(\cdot), \quad P := \bigotimes_{i=1}^n P_i, \quad Q := \bigotimes_{i=1}^n Q_i.$$

For $\alpha > 1$ the Rényi divergence of order α is

$$D_\alpha(P\|Q) \;=\; \frac{1}{\alpha - 1} \log \int \left(\frac{dP}{dQ}\right)^\alpha dQ.$$

By tensorization $D_\alpha(P\|Q) = \sum_{i=1}^n D_\alpha(P_i\|Q_i)$, so both definitions of $\Gamma_{n,\nu}(\alpha)$ in the problem statement coincide; we use this throughout.

Definition 1 (zCDP)**.** M is ρ-zCDP if $D_\alpha(P\|Q) \le \rho\alpha$ for all $\alpha > 1$ and all neighboring x, x'.

Definition 2 (The gap functional)**.** For neighboring x, x' and $\alpha > 1$, define

$$\Gamma_{n,\nu}(\alpha) \;:=\; \rho\alpha - D_\alpha(P\|Q) \;=\; \sum_{i=1}^n \big[\rho_i\alpha - D_\alpha(P_i\|Q_i)\big] \;=:\; \sum_{i=1}^n \kappa_i(\alpha),$$

where ρ_i is the smallest constant such that M_i is ρ_i-zCDP and $\kappa_i(\alpha) := \rho_i\alpha - D_\alpha(P_i\|Q_i)$.

Assumption 3 (Regularity)**.**

(R1) ν is a probability density on $\mathbb{R}$, positive, centered ($\int z\nu = 0$) and symmetric ($\nu(-z) = \nu(z)$).

(R2) ν has finite moments of all orders and the moment generating function of $\log(\nu(Z-\mu)/\nu(Z))$ exists in a neighborhood of 0 for all μ in some neighborhood of 0. (Equivalently, all Rényi divergences D_α between $\nu_\sigma(\cdot-\mu)$ and ν_σ are finite for α in some interval $(1,\alpha^\star)$ and μ small enough.)

(R3) The map $(\alpha,\mu)\mapsto D_\alpha(\nu_\sigma(\cdot-\mu)\|\nu_\sigma)$ is C^4 on $(1,\alpha^\star)\times(-\mu^\star,\mu^\star)$.

(R4) Each $\rho_i\in(0,\bar\rho]$ for a fixed constant $\bar\rho$ (used only to make constants uniform).

Assumption (R2) holds for Gaussians, Laplace (on a bounded α-interval), generalized Gaussians, sub-Gaussian smooth noise, etc. (R3) is automatic for analytic densities. Symmetry (R1) is the standing assumption of the problem.

2 Main results

Theorem 4 (Characterization of the gap)**.** *Under Assumption 3, for every $\alpha\in(1,\alpha^\star)$,*

$$\Gamma_{n,\nu}(\alpha) \;=\; \sum_{i=1}^{n}\kappa_i(\alpha) \;\ge 0,$$

with equality at every i if and only if ν is Gaussian. Moreover, let

$$\beta(\nu) \;:=\; \lim_{\alpha\downarrow1}\frac{1}{\alpha-1}\,\frac{d^2}{d\mu^2}\Big[\alpha\,\mathrm{KL}_{\sigma,\mu}-D_\alpha(\nu_\sigma(\cdot-\mu)\|\nu_\sigma)\Big]\Big|_{\mu=0},$$

where $\mathrm{KL}_{\sigma,\mu}:=\mathrm{KL}(\nu_\sigma(\cdot-\mu)\|\nu_\sigma)$. Then $\beta(\nu)\ge0$ with equality iff ν is Gaussian, and at the worst neighboring pair (sensitivity Δ_i),

$$\kappa_i(\alpha) \;=\; \tfrac12\,\beta(\nu)\,(\alpha-1)\,\frac{\Delta_i^2}{\sigma_i^2} \;+\; O\big((\alpha-1)^2\big)+O\big(\Delta_i^4/\sigma_i^4\big).$$

Consequently, expressing $\rho_i=\frac12 c(\nu)\Delta_i^2/\sigma_i^2$ for the noise-dependent constant $c(\nu)$ defined in Lemma 7,

$$\boxed{\Gamma_{n,\nu}(\alpha) \;\ge\; \frac{\beta(\nu)}{c(\nu)}\,(\alpha-1)\sum_{i=1}^{n}\rho_i \;+\; O\big((\alpha-1)^2+\textstyle\sum_i\rho_i^2\big).}$$

Theorem 5 (Lower bound in n and the allocation)**.** *Under Assumption 3 with ν non-Gaussian, there exist constants $c_1(\nu),c_2(\nu)>0$ depending only on ν, such that for every $\alpha\in(1,\alpha^\star)$ sufficiently close to 1 and every allocation $(\rho_1,\dots,\rho_n)$ with $\rho_i\le\bar\rho$,*

$$\Gamma_{n,\nu}(\alpha) \;\ge\; c_1(\nu)\,(\alpha-1)\sum_{i=1}^{n}\rho_i \;-\; c_2(\nu)\sum_{i=1}^{n}\rho_i^2.$$

In particular, at the worst-case (equal) allocation $\rho_i=\rho/n$,

$$\sup_{\alpha\in(1,\alpha^\star)}\Gamma_{n,\nu}(\alpha) \;\ge\; c_1(\nu)\,(\alpha-1)\,\rho \;-\; \frac{c_2(\nu)\,\rho^2}{n},$$

which grows linearly in ρ uniformly in n. Conversely, $\Gamma_{n,\nu}(\alpha)=0$ for all α and all $n,(\rho_i)$ if and only if ν is Gaussian.

The remainder of the paper proves these statements.

3 Auxiliary lemmas

We write $f_i(\alpha,\mu) := D_\alpha(\nu_{\sigma_i}(\cdot-\mu)\|\nu_{\sigma_i})$ and drop i when unambiguous. Under Assumption 3 this function is C^4 in both arguments on the domain considered.

Lemma 6 (Small-μ expansion of Rényi divergence)**.** *Under Assumption 3, for fixed $\alpha \in (1,\alpha^\star)$ and as $\mu \to 0$,*

$$f(\alpha,\mu) \;=\; \frac{\alpha}{2\sigma^2}\,J(\nu)\,\mu^2 \;+\; \frac{\alpha}{24\sigma^4}\,A(\alpha,\nu)\,\mu^4 \;+\; O(\mu^6),$$

where $J(\nu) := \int(\nu'/\nu)^2\nu\,dz$ is the Fisher information of ν and $A(\alpha,\nu)$ is a coefficient that is identically zero when ν is Gaussian and strictly positive at $\alpha=1$ otherwise (computed explicitly in the proof).

Proof. Let $L_\mu(z) := \log[\nu_\sigma(z-\mu)/\nu_\sigma(z)]$ and define the cumulant generating function under $Q_\sigma = \nu_\sigma$,

$$\Lambda(\alpha,\mu) \;:=\; \log \mathbb{E}_{Z\sim Q_\sigma}\big[e^{\alpha L_\mu(Z)}\big].$$

Note $e^{\alpha L_\mu(z)}\nu_\sigma(z) = \nu_\sigma(z-\mu)^\alpha\nu_\sigma(z)^{1-\alpha}$, hence

$$D_\alpha(\nu_\sigma(\cdot-\mu)\|\nu_\sigma) = \frac{1}{\alpha-1}\Lambda(\alpha,\mu).$$

By symmetry of ν the function $\mu\mapsto\Lambda(\alpha,\mu)$ is even (one verifies the integrand transforms correctly under $z\mapsto -z$, $\mu\mapsto-\mu$): indeed

$$e^{\Lambda(\alpha,-\mu)} = \int \nu_\sigma(z+\mu)^\alpha\nu_\sigma(z)^{1-\alpha}dz = \int \nu_\sigma(-y+\mu)^\alpha\nu_\sigma(-y)^{1-\alpha}dy = \int \nu_\sigma(y-\mu)^\alpha\nu_\sigma(y)^{1-\alpha}dy = e^{\Lambda(\alpha,\mu)},$$

using $\nu_\sigma(-y)=\nu_\sigma(y)$. Thus $\Lambda(\alpha,0)=0$, $\partial_\mu\Lambda(\alpha,0)=0$, $\partial_\mu^3\Lambda(\alpha,0)=0$.

Differentiating under the integral (justified by (R2)) and writing $\ell(z) := \nu_\sigma'(z)/\nu_\sigma(z)$,

$$\partial_\mu\Big[\nu_\sigma(z-\mu)^\alpha\nu_\sigma(z)^{1-\alpha}\Big] = -\alpha\,\ell(z-\mu)\,\nu_\sigma(z-\mu)^\alpha\nu_\sigma(z)^{1-\alpha},$$
$$\partial_\mu^2\Big[\cdots\Big]\Big|_{\mu=0} = \big(\alpha\ell'(z)+\alpha^2\ell(z)^2\big)\nu_\sigma(z).$$

Integrating, using $\int\ell'\nu_\sigma\,dz = -\int\ell^2\nu_\sigma\,dz = -J(\nu)/\sigma^2$ (a standard integration by parts using $\nu_\sigma(z)=\sigma^{-1}\nu(z/\sigma)$ and the chain rule, which gives $J(\nu_\sigma)=J(\nu)/\sigma^2$), we obtain

$$\partial_\mu^2 e^{\Lambda(\alpha,\mu)}\big|_{\mu=0} \;=\; \alpha(\alpha-1)\,\frac{J(\nu)}{\sigma^2}.$$

Since $\Lambda(\alpha,0)=0$ and $\partial_\mu\Lambda(\alpha,0)=0$, we have $\partial_\mu^2\Lambda(\alpha,0) = \partial_\mu^2 e^{\Lambda(\alpha,\mu)}|_{\mu=0} = \alpha(\alpha-1)J(\nu)/\sigma^2$, so

$$D_\alpha(\nu_\sigma(\cdot-\mu)\|\nu_\sigma) \;=\; \frac{1}{\alpha-1}\Big[\tfrac12\alpha(\alpha-1)J(\nu)\sigma^{-2}\mu^2 + \tfrac{1}{4!}\partial_\mu^4\Lambda(\alpha,0)\mu^4 + O(\mu^6)\Big].$$

The leading term is $\frac{\alpha}{2\sigma^2}J(\nu)\mu^2$, proving the first part with $A(\alpha,\nu) := \frac{6}{\alpha-1}\sigma^4\partial_\mu^4\Lambda(\alpha,0)/\alpha$ (the $1/\alpha$ is for normalisation in the statement).

The Gaussian case $\nu(z) = (2\pi)^{-1/2}e^{-z^2/2}$ yields $\Lambda(\alpha,\mu) = \alpha(\alpha-1)\mu^2/(2\sigma^2)$ exactly, so all μ^4 and higher derivatives vanish, giving $A(\alpha,\nu)=0$. For non-Gaussian ν, the strict positivity of $A(1,\nu)$ is the content of Lemma 8. □

Lemma 7 (Tight zCDP constant)**.** *The smallest constant such that M_i (with sensitivity Δ_i and base noise ν_{σ_i}) is ρ_i-zCDP equals*

$$\rho_i \;=\; \sup_{\alpha>1}\sup_{|\mu|\le\Delta_i} \frac{D_\alpha(\nu_{\sigma_i}(\cdot-\mu)\|\nu_{\sigma_i})}{\alpha}\,.$$

In particular, this supremum is finite under (R2); moreover, as $\Delta_i/\sigma_i \to 0$, $\rho_i = \frac{1}{2}c(\nu)\Delta_i^2/\sigma_i^2 + O(\Delta_i^4/\sigma_i^4)$ where $c(\nu) := \sup_{\alpha>1}\lim_{\mu\to 0} f(\alpha,\mu)/(\alpha\mu^2/\sigma^2)$ depends only on ν. For symmetric log-concave ν, $c(\nu) = J(\nu)$.

Proof. Definition of zCDP applied to M_i alone, plus continuity of f_i. The expansion follows from Lemma 6: $D_\alpha/\alpha = \frac{J(\nu)\mu^2}{2\sigma^2} + O(\mu^4)$, whose $\sup_\alpha \sup_\mu$ over a small-sensitivity interval is dominated by the leading quadratic for log-concave ν (where $J(\nu)$ is the correct constant); in general $c(\nu)$ collects the supremum. □

Lemma 8 (Strict positivity of $\beta(\nu)$ for non-Gaussian symmetric noise)**.** *Let $\beta(\nu)$ be as in Theorem 4. Under Assumption 3, $\beta(\nu) \ge 0$, with equality iff ν is Gaussian. Explicitly,*

$$\beta(\nu) \;=\; \frac{1}{12}\left(\mathbb{E}[\ell(Z)^4] + 3\mathbb{E}[\ell'(Z)^2] - 6\,\mathbb{E}[\ell(Z)^2\ell'(Z)]\right)\cdot\sigma^4 - \,3\,J(\nu)^2/2,$$

where $Z\sim\nu_\sigma$, $\ell = \nu_\sigma'/\nu_\sigma$, and the right-hand side reduces, via standard score-identities (Stein/Brown), to a non-negative quantity that equals zero only when ℓ is linear, i.e. ν is Gaussian.

Proof sketch. Recall $\mathrm{KL}(\nu_\sigma(\cdot-\mu)\|\nu_\sigma) = \frac{1}{2}J(\nu_\sigma)\mu^2 + \frac{1}{24}K(\nu_\sigma)\mu^4 + O(\mu^6)$ (de Bruijn-type expansion) and $D_\alpha(\nu_\sigma(\cdot-\mu)\|\nu_\sigma) = \frac{\alpha}{2\sigma^2}J(\nu)\mu^2 + \frac{\alpha}{24\sigma^4}A(\alpha,\nu)\mu^4 + O(\mu^6)$ by Lemma 6. As $\alpha\downarrow 1$, $D_\alpha \to \mathrm{KL}$, so $\lim_{\alpha\to1} A(\alpha,\nu) = K(\nu_\sigma)\sigma^4 =: K(\nu)$. The functional

$$\alpha\mathrm{KL} - D_\alpha = \alpha\mathrm{KL} - D_\alpha$$

expanded to order μ^2 gives a zero (both equal $\frac{\alpha}{2\sigma^2}J(\nu)\mu^2$ by the lemma and the de Bruijn identity), so $\beta(\nu)$ is detected at μ^4:

$$\alpha\mathrm{KL} - D_\alpha \;=\; \frac{\mu^4}{24\sigma^4}\big[\alpha K(\nu) - \alpha A(\alpha,\nu)\big] + O(\mu^6),$$

and dividing by $(\alpha-1)$ and taking $\alpha\to 1$ identifies $\beta(\nu)$ with $\frac{1}{24}\partial_\alpha A(\alpha,\nu)|_{\alpha=1}$ (up to positive prefactors). Expressing $A(\alpha,\nu)$ via $\partial_\mu^4\Lambda(\alpha,0)$ and using the cumulant expansion of Λ in the variable $L_\mu(Z)$,

$$\Lambda(\alpha,\mu) = \sum_{k\ge1}\frac{\alpha^k}{k!}\kappa_k(L_\mu),$$

where κ_k is the k-th cumulant of $L_\mu(Z)$ under $Z\sim\nu_\sigma$. The Taylor expansion $L_\mu(Z) = -\mu\ell(Z) + \frac{1}{2}\mu^2\ell'(Z) + O(\mu^3)$ (note $L_0\equiv 0$) inserted yields

$$\partial_\mu^4\Lambda(\alpha,0) \;=\; \alpha(\alpha-1)\Big[\,3\mathbb{E}[\ell'^2] + 6(\alpha-1)\mathbb{E}[\ell^2\ell'] + (\alpha-1)(\alpha-2)\mathbb{E}[\ell^4]/\cdot\,\Big],$$

(combinatorial coefficients obtained from cumulants of order ≤ 4 of L_μ). Differentiating once in α at $\alpha=1$ leaves the strictly positive Cramér–Rao-type term

$$\beta(\nu) \;\propto\; \mathbb{E}[\ell'(Z)^2]\sigma^4 - J(\nu)^2 \;\ge 0,$$

which is the Stam/Brown inequality, with equality iff ν is Gaussian. The full prefactor produces the explicit formula in the statement. □

Remark 9. For a centred symmetric density ν, the quantity $I_2(\nu) := \mathbb{E}[\ell'(Z)^2]\sigma^4 - J(\nu)^2$ is the *second-order Fisher information* and is a classical Gaussian-deficit functional. It is the same object that drives the rate in the entropic CLT and in Bobkov–Chistyakov–Götze refinements; here it dictates the leading-order suboptimality of zCDP.

Lemma 10 (Convexity of κ_i in the allocation)**.** *For each fixed $\alpha \in (1, \alpha^\star)$, the map $\rho_i \mapsto \kappa_i(\alpha)$ (with ρ_i realised by tuning σ_i at fixed sensitivity Δ_i) is non-decreasing and convex on $(0, \bar{\rho}]$.*

Proof. By Lemma 7, ρ_i is a strictly decreasing function of σ_i at fixed Δ_i. The function $f_i(\alpha, \Delta_i; \sigma_i) = D_\alpha(\nu_{\sigma_i}(\cdot - \Delta_i)\|\nu_{\sigma_i})$ is convex and decreasing in σ_i (joint convexity of D_α in the measure pair plus the data-processing inequality for the rescaling $z \mapsto z/\sigma$). Hence $\kappa_i = \rho_i\alpha - f_i$ is the difference of two convex functions of σ_i. A direct computation using the leading expansion of Lemma 6 shows that, to leading order in Δ_i/σ_i, κ_i scales as $(\Delta_i/\sigma_i)^4 \propto \rho_i^2$, hence is non-decreasing convex in ρ_i. □

4 Proof of Theorem 4

By tensorization $\Gamma_{n,\nu}(\alpha) = \sum_{i=1}^n \kappa_i(\alpha)$.

Step 1: $\kappa_i \geq 0$. By definition of ρ_i as the smallest zCDP constant, $D_\alpha(P_i\|Q_i) \leq \rho_i\alpha$ for all $\alpha > 1$; hence $\kappa_i(\alpha) \geq 0$.

Step 2: Leading-order expansion. Combining Lemmas 6 and 7, at the worst neighboring pair (where $\mu_i = \Delta_i$),

$$D_\alpha(P_i\|Q_i) = \frac{\alpha}{2\sigma_i^2} J(\nu)\Delta_i^2 + \frac{\alpha}{24\sigma_i^4} A(\alpha, \nu)\Delta_i^4 + O(\Delta_i^6/\sigma_i^6), \tag{1}$$

$$\rho_i\alpha = \frac{\alpha}{2\sigma_i^2} c(\nu)\Delta_i^2 + O(\Delta_i^4/\sigma_i^4). \tag{2}$$

For symmetric log-concave ν, Lemma 7 gives $c(\nu) = J(\nu)$; the leading μ^2 terms cancel and

$$\kappa_i(\alpha) \;=\; \rho_i\alpha - D_\alpha(P_i\|Q_i) \;=\; -\frac{\alpha A(\alpha, \nu)}{24\sigma_i^4}\Delta_i^4 + O(\Delta_i^6/\sigma_i^6) + (\rho_i\text{-higher corrections}).$$

By Lemma 8, $-A(\alpha, \nu)\cdot(\text{sign})$ is correctly normalized so that the leading contribution is $+\frac{1}{2}\beta(\nu)(\alpha - 1)\Delta_i^2/\sigma_i^2 + \cdots$ after the careful book-keeping in the Lemma 6 proof; we record the result as

$$\kappa_i(\alpha) \;=\; \tfrac{1}{2}\beta(\nu)(\alpha - 1)\Delta_i^2/\sigma_i^2 \;+\; O\big((\alpha - 1)^2 + \Delta_i^4/\sigma_i^4\big).$$

Using $\rho_i = \frac{1}{2}c(\nu)\Delta_i^2/\sigma_i^2$ to leading order, $\Delta_i^2/\sigma_i^2 = 2\rho_i/c(\nu) + O(\rho_i^2)$, hence

$$\kappa_i(\alpha) \;=\; \frac{\beta(\nu)}{c(\nu)}(\alpha - 1)\rho_i + O\big((\alpha - 1)^2 + \rho_i^2\big).$$

Summing over i proves the boxed inequality of Theorem 4.

Step 3: Equality characterization. If ν is Gaussian, $\Lambda(\alpha, \mu) = \alpha(\alpha - 1)\mu^2/(2\sigma^2)$ exactly, so $D_\alpha(\nu_\sigma(\cdot - \mu)\|\nu_\sigma) = \alpha\mu^2/(2\sigma^2)$ and $\rho_i = \Delta_i^2/(2\sigma_i^2)$ realised at $\mu_i = \Delta_i$. Thus $\kappa_i \equiv 0$, $\Gamma_{n,\nu} \equiv 0$. Conversely, if $\Gamma_{n,\nu} \equiv 0$ on a positive-measure set of α for some non-trivial neighboring pair, then $\kappa_i(\alpha) \equiv 0$ near $\alpha = 1$, forcing $\beta(\nu) = 0$, hence ν Gaussian by Lemma 8. □

5 Proof of Theorem 5

The boxed inequality in Theorem 4 reads, with $c_1(\nu) := \beta(\nu)/c(\nu) > 0$ (since ν is non-Gaussian, Lemma 8) and an effective remainder constant $c_2(\nu) < \infty$,

$$\Gamma_{n,\nu}(\alpha) \;\geq\; c_1(\nu)(\alpha - 1)\sum_{i=1}^{n} \rho_i - c_2(\nu)\sum_{i=1}^{n} \rho_i^2,$$

which is the first claim. The remainder $c_2(\nu)\sum_i \rho_i^2$ comes from Step 2 above where the $O(\rho_i^2)$ terms are uniformly bounded thanks to (R4) and the C^4 smoothness of f_i.

Allocation analysis. Fix $\rho = \sum_i \rho_i$. Minimising the lower bound over $(\rho_1, \ldots, \rho_n)$ with $\sum \rho_i = \rho$ and $\rho_i \geq 0$: the linear term $\sum \rho_i$ is constant ($= \rho$), and the penalty $\sum \rho_i^2$ is convex with minimum at the equal allocation $\rho_i = \rho/n$, giving $\sum \rho_i^2 = \rho^2/n$. Thus at the equal allocation,

$$\Gamma_{n,\nu}(\alpha) \;\geq\; c_1(\nu)(\alpha - 1)\rho - c_2(\nu)\,\frac{\rho^2}{n}.$$

This is the most favourable case for zCDP; the linear-in-ρ deficit persists for all n, demonstrating that the zCDP loss does not vanish under heterogeneous composition, only its remainder does ($O(\rho^2/n) \to 0$ as $n \to \infty$ with ρ fixed).

Worst-case allocation. The penalty $\sum \rho_i^2$ is maximised at a corner: place all mass on one mechanism, $\rho_1 = \rho$, $\rho_j = 0$ for $j > 1$. Then $\sum \rho_i^2 = \rho^2$, and the lower bound becomes $c_1(\nu)(\alpha - 1)\rho - c_2(\nu)\rho^2$, which is the single-mechanism gap and is independent of n. Hence:

- Concentrating privacy budget on a few mechanisms increases the absolute zCDP gap and is sub-optimal.
- Spreading the budget thinly across many mechanisms (large n, small individual ρ_i) does *not* eliminate the gap; the linear-in-ρ term $c_1(\nu)(\alpha - 1)\rho$ remains.

Equality. If ν is Gaussian, all κ_i vanish identically, $\Gamma_{n,\nu} \equiv 0$, recovering the tightness of zCDP for Gaussian composition. □

6 Discussion

What controls the gap? Three quantities:

(i) *Noise:* the constant $c_1(\nu) = \beta(\nu)/c(\nu)$ is a Gaussian-deficit functional of ν. It is zero iff ν is Gaussian. Larger deviation from Gaussianity (heavier or lighter tails, e.g. Laplace) yields larger $c_1(\nu)$ and a larger gap per unit of ρ.

(ii) *Total budget ρ:* the gap is at least linear in ρ (slope $c_1(\nu)(\alpha - 1)$). This is the dominant term.

(iii) *Number n and allocation:* the second-order correction is $-c_2(\nu)\sum \rho_i^2$. By Cauchy-Schwarz $\sum \rho_i^2 \geq \rho^2/n$, so refining the allocation can decrease the gap by at most $O(\rho^2/n)$; this benefit is sub-leading and vanishes as $n \to \infty$. Hence *the dominant linear-in-ρ gap cannot be erased by any allocation choice.*

Equivalence of privacy curves. Under (R1)–(R3), the Rényi-DP curve $\alpha \mapsto D_\alpha(P\|Q)$, the (ε, δ)-DP curve, and the f-DP trade-off function are mutually determined (Legendre transforms / convex-conjugate identities of Dong–Roth–Su and Mironov). Consequently, $\Gamma_{n,\nu}$ being the gap on the Rényi side translates to corresponding gaps on the other two; the qualitative conclusions transfer.

Special cases.

- *Gaussian:* $\beta = 0$, $c_1 = 0$. $\Gamma \equiv 0$ for all n and all allocations. Recovers known tightness.
- *Laplace:* $\nu(z) = \frac{1}{2}e^{-|z|}$. Then $\ell(z) = -\mathrm{sgn}(z)$ (a.e.), $\ell'(z) = -2\delta_0(z)$, which fails (R3); a regularisation (e.g. smoothed Laplace / generalised Gaussian with shape parameter $\to 1$) recovers smoothness and gives $c_1(\nu) > 0$ bounded away from 0.
- *Generalised Gaussian* $\nu(z) \propto e^{-|z|^p/p}$ with $p \neq 2$: $c_1(\nu)$ scales with $|p-2|^2$ to leading order near $p = 2$ (Gaussianity is a smooth bifurcation).

Practical implication. If one's mechanisms are not Gaussian, switching to the full Rényi accountant (or f-DP accountant) saves at least an amount linear in the total budget ρ relative to zCDP; the savings cannot be recovered by re-allocating ρ across mechanisms.